BUS105

Examination - January Semester 2025

Statistics

Thursday, 17 April 2025 **10:00 am - 12:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **4** questions and comprises **10** printed pages (including cover page).
2. You must answer **4** questions.
3. All answers must be written in the answer book.
4. This is **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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(Full marks: 100)

Question 1

Question 1a

The data on Singapore employment, extracted from the Ministry of Manpower, for 2013 and 2023, is shown in ***Table 1***. Explain the changes in employment by industry over this period. In your answer, provide data and comments accordingly. Please limit your answer to 200 words.

Table 1: *Singapore Employment for 2013 and 2023 (in Thousands)*

Singapore Employment	**Dec-23**	**Dec-13**
Total	3,999.8	3,493.6
Manufacturing	485.2	539.9
Construction	526.2	479.5
Services	2,964.0	2,447.9
Others	24.4	26.2

(6 marks)

Question 1b

Without using Excel, define the mean, variance and standard deviation of the number of times a communication (x) is made between two banks operating in Singapore and Hong Kong, respectively. Show your workings with the appropriate formulae.

Number of times a communication (x) is made between two banks	Proportion (Probability, p(x))
0	0.2
1	0.4
2	0.4

(6 marks)

Question 1c

The family expenditure per month in Singapore for a household of four for 2024 is estimated to have a population mean of $5,500 and a standard deviation of $350. Without using Excel and assuming the data is normally distributed, define the probability that the family expenditure is higher than $5,700. Show your workings including formulae and graphical illustration.

(5 marks)

Question 1d

If 25 families are randomly selected for interview on their family expenditure, report the probability that the mean expenditure is between $5,350 and $5,705 for this sample, based on a population mean of $5,500 and standard deviation of $350, assuming the sample is from a Normal Distribution.

(8 marks)

Question 2

Question 2a

SuperGolf wanted to estimate the mean expenditure for the manufacturing of its golf clubs. A random group of 150 golf clubs were selected and their average expenditure is $400. Based on historical data, the population standard deviation is known to be $8.

Without using Excel, identify the 90% Confidence Interval for the mean expenditure for the manufacturing of golf clubs by SuperGolf. Show your workings including formulae and graphical illustration.

(10 marks)

Question 2b

SuperGolf reports the mean expenditure to manufacture golf clubs is equal to $400. A comparison from other golf club manufacturers shows this amount to be larger than most other manufacturers, so SuperGolf believes there is a need to reduce the expenditure. To evaluate the effect of expenditure reduction, the Manager at SuperGolf selected a random sample of 10 golf clubs and the manufacturing expenditure is indicated in ***Table 2***.

At the 0.05 significance level, is it reasonable to conclude that the mean expenditure is less than $400?

Use the 5-step procedure to execute an appropriate hypothesis test for SuperGolf's manufacturing expenditure. Show your workings including formulae and graphical illustration.

Table 2: *Golf Club Manufacturing Expenditure*

Golf Club	Expenditure
1	402
2	392
3	398
4	405
5	407
6	387
7	404
8	406
9	395
10	403

(15 marks)

Question 3

Question 3a

SuperGolf's manufacturing expenditure of golf clubs is shown under "Iron" and "Hybrid" in ***Table 3***.

At the 0.05 significance level, using the 5-step procedure, can you conclude there is a difference in the manufacturing expenditure between "Iron" and "Hybrid" according to the Excel analysis as shown in ***Table 4***?

Table 3: *Golf Club Manufacturing Expenditures*

Golf Club	Iron	Hybrid
1	402	385
2	392	410
3	398	405
4	405	375
5	407	390
6	387	345
7	404	400
8	406	395
9	395	400
10	403	388

Table 4: *Excel Analysis*

t-Test: Paired Two Sample for Means

	Iron	*Hybrid*
Mean	399.9	389.3
Variance	44.54444444	347.1222222
Observations	10	10
Pearson Correlation	0.288885147	
Hypothesized Mean Difference	0	
Df	9	
t Stat	1.874353696	
P(T<=t) one-tail	0.046823077	
t Critical one-tail	1.833112933	
P(T<=t) two-tail	0.093646155	
t Critical two-tail	2.262157163	

(11 marks)

Question 3b

SuperGolf has two sub-contractors: NewGolf and BigGolf. A sample of five (5) golf clubs that are manufactured by the three (3) companies is shown in ***Table 5***.

At a 0.05 significance level, using the 5-step procedure, execute an analysis of variance (ANOVA) procedure, as shown in ***Table 6***, to determine if there is a difference in the mean expenditure for the three (3) companies. You should include a graphical illustration as part of your ANOVA results.

Table 5: *Golf Club Manufacturing Expenditures*

Golf Club	SuperGolf Expenditure_1	NewGolf Expenditure_2	BigGolf Expenditure_3
1	402	385	395
2	392	410	380
3	398	405	375
4	405	375	400
5	407	390	350

Table 6: *ANOVA Table*

ANOVA

Source of Variation	*SS*	*Df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	1104.133333	2	552.0666667	2.625971143	0.113255551	3.885293835
Within Groups	2522.8	12	210.2333333			
Total	3626.933333	14				

(14 marks)

Question 4

The sale of SuperGolf's golf clubs across cities in the US is dependent on the city's population, the number of golf players, advertising and whether there is a store in the city ("1" has a store; "0" no store). A random sample of 15 cities is selected (see ***Table 7***).

Table 7: *Golf Clubs Sales by SuperGolf across Cities in the US*

Sales ($,000)	Population (000,000)	Number of Golf Players (,000)	Advertising ($,000)	Store
12.10	8.65	5.8	5.9	0
10.50	7.18	6.2	3.2	1
8.50	6.25	5.5	5.8	0
10.30	9.85	6.8	4.9	0
8.80	5.55	6.2	4.1	1
9.96	4.40	6.3	3.7	0
9.83	7.35	0.5	4.0	0
10.12	8.15	5.5	3.2	1
10.70	6.48	5.8	4.3	1
10.45	7.84	7.1	4.2	0
11.32	6.64	7.5	3.0	0
11.87	8.83	8.2	2.7	1
11.91	4.27	7.8	5.8	1
12.60	7.65	6.2	5.0	0
12.80	8.75	7.1	4.6	1

The Excel multiple regression outputs are shown in ***Table 8***.

Table 8*: Excel Multiple Regression Outputs for SuperGolf Sales*

Regression Statistics	
Multiple R	0.544050636
R Square	0.295991094
Adjusted R Square	0.014387532
Standard Error	1.284735795
Observations	15

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	4	6.939499361	1.73487484	1.051091441	0.428878869
Residual	10	16.50546064	1.650546064		
Total	14	23.44496			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*
Intercept	6.001494908	2.625027196	2.286260088	0.045300927	0.152569826
Population (000,000)	0.282316176	0.215087087	1.312566829	0.218654272	-0.19692772
Number of Golf Players (,000)	0.275533041	0.202176033	1.362837314	0.202833003	-0.174943232
Advertising ($,000)	0.216547489	0.347196462	0.62370304	0.546785255	-0.557054437
Store	0.265733621	0.729572443	0.364231988	0.723268426	-1.359855084

Question 4a

Illustrate the multiple regression equation using all four independent variables and clearly define the variables (e.g., X, Y) in your regression equation.

(6 marks)

Question 4b

Interpret the coefficients between the dependent variable and each of the independent variables.

(8 marks)

Question 4c

Determine the adjusted coefficient of multiple determination.

(2 marks)

Question 4d

Execute a global hypothesis test at 0.05 (Level of Significance (α)) to determine if at least one regression coefficient is non-zero.

(9 marks)

----- END OF PAPER -----